

PROJECT TITLE: Movie Plot Compressor

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Abstract:

The Movie Plot Compressor is an automated NLP pipeline designed to transform extensive Wikipedia movie plot descriptions into concise, one-paragraph summaries. By leveraging the BART (Bidirectional and Auto-Regressive Transformers) model, specifically the facebook/bart-large-cnn architecture the system effectively distills complex narratives while maintaining chronological coherence and thematic accuracy. The project bridges the gap between raw web data and high-quality content summarization, providing users with instant, digestible insights into cinematic works.

1. The Objective

The core challenge was to programmatically ingest long-form text from Wikipedia and condense it into a meaningful blurb. The solution required a multi-stage approach:

- Data Acquisition: Interfacing with the Wikipedia API to fetch specific movie pages.

- Information Extraction: Parsing raw HTML/text to isolate the "Plot" or "Synopsis" sections while discarding irrelevant metadata or citations.

- Neural Summarization: Utilizing a pre-trained Transformer model capable of understanding context to generate a natural language summary.

2. System Architecture & Components

The notebook is structured into a logical execution flow to ensure modularity and reliability:

A. Data Scraping & Pre-processing

- Wikipedia Integration: Uses the wikipedia library to search for and retrieve page content.

- Metadata Extraction: Employs BeautifulSoup and requests to scrape the movie's "Infobox" for specific details like Genre and Release Year.

- Text Cleaning: Implements regular expressions to strip Wikipedia citation brackets and isolate the plot text from subsequent sections

B. The Summarization Engine

- Model Selection: The system utilizes BART, a denoising autoencoder for pre-training sequence-to-sequence models. It is particularly effective for summarization tasks because it combines the bidirectional encoder of BERT with the left-to-right decoder of GPT.

- Tokenization: The BartTokenizer manages text truncation to 1024 tokens to fit within the model's architectural constraints.

- Generation Parameters: The summarize_plot function uses Beam Search (num_beams=4) and a length penalty to ensure the output is neither too brief nor repetitive, maintaining a target length of 100–200 words.

3. Key Features

- Automated Search: If a direct movie title match isn't found, the system intelligently searches for the title with the suffix "film" to improve accuracy.

-Structured Output: Beyond the paragraph summary, the system formats the result into a 10-Line Format for even faster scannability.

-Error Handling: Includes checks for Disambiguation Errors (where multiple pages exist for one title) and missing model dependencies.

4. Technical Stack

-Language: Python 3.11

-Libraries: transformers (Hugging Face), torch (PyTorch), beautifulsoup4, wikipedia, re.

-Model: facebook/bart-large-cnn

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Interactive Input (Keep Entering Movies)
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```
while True:
    movie_name = input("\nEnter movie name (or 'quit' to exit): ")

    if movie_name.lower() in ['quit', 'exit', 'q']:
        print("\nGoodbye!")
        break

    if movie_name.strip():
        get_movie_summary(movie_name)
    else:
        print("Please enter a valid movie name.")
```

✓ 6m 2.1s Python

Found: The Truman Show

📄 MOVIE INFORMATION

Title: The Truman Show
Genre: Unknown
Year: 1998

Plot length: 2000 characters

Generating summary (this may take 10-30 seconds)...

SUMMARY

The Truman Show is a 1998 American comedy-drama film written and co-produced by Andrew Niccol, and directed by Peter Weir. The film depicts the story

KEY POINTS (10-Line Format)

1. The Truman Show is a 1998 American comedy-drama film written and co-produced by Andrew Niccol, and directed by Peter Weir.
2. The film depicts the story of Truman Burbank, a man who is unaware that he is living his entire life on a colossal soundstage.
3. The majority of filming took place at Seaside, Florida, a master-planned community located in the Florida Panhandle.
4. The Truman Show held its world premiere in Los Angeles on June 1, 1998, and was released in North America on June 5.

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📽 MOVIE INFORMATION

Title: Pulp Fiction
Genre: Unknown
Year: 1994

Plot length: 2000 characters

Generating summary (this may take 10-30 seconds)...

SUMMARY

Pulp Fiction is a 1994 American independent crime film written and directed by Quentin Tarantino from a story he conceived with Roger Avary. The film

KEY POINTS (10-Line Format)

1. Pulp Fiction is a 1994 American independent crime film written and directed by Quentin Tarantino from a story he conceived with Roger Avary.
 2. The film stars John Travolta, Samuel L.
 3. Jackson, Bruce Willis, Tim Roth, Ving Rhames, and Uma Thurman.
 4. It won the Palme d'Or at the 1994 Cannes Film Festival and was a major critical and commercial success.
 5. It was nominated for seven awards at the 67th Academy Awards, including Best Picture, and won Best Original Screenplay.
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